

Module 2: Contextual Analysis

Methods for Analyzing Challenges & Opportunities - People and Systems,
Patterns and Priorities, Problem Framing.

Contextual Analysis: Consolidating and Interpreting Work Activity Data

Contextual Analysis

- Contextual Analysis is a method used to interpret and synthesize data collected from real-world environments to understand the context in which users perform their activities.
- It helps researchers and designers identify challenges and opportunities, providing insights that guide problem-solving and solution design.

Analyzing Challenges & Opportunities

- **Challenges** refer to the barriers or difficulties users face while performing tasks.
- **Opportunities** refer to areas where interventions, improvements, or innovations can enhance user experience or performance.
- Contextual analysis helps **identify both by looking at patterns in how people interact with systems, tools, and their environment.**

Methods for Analyzing Challenges & Opportunities:-

- People and Systems
- Patterns and Priorities
- Problem Framing.

1. People and Systems

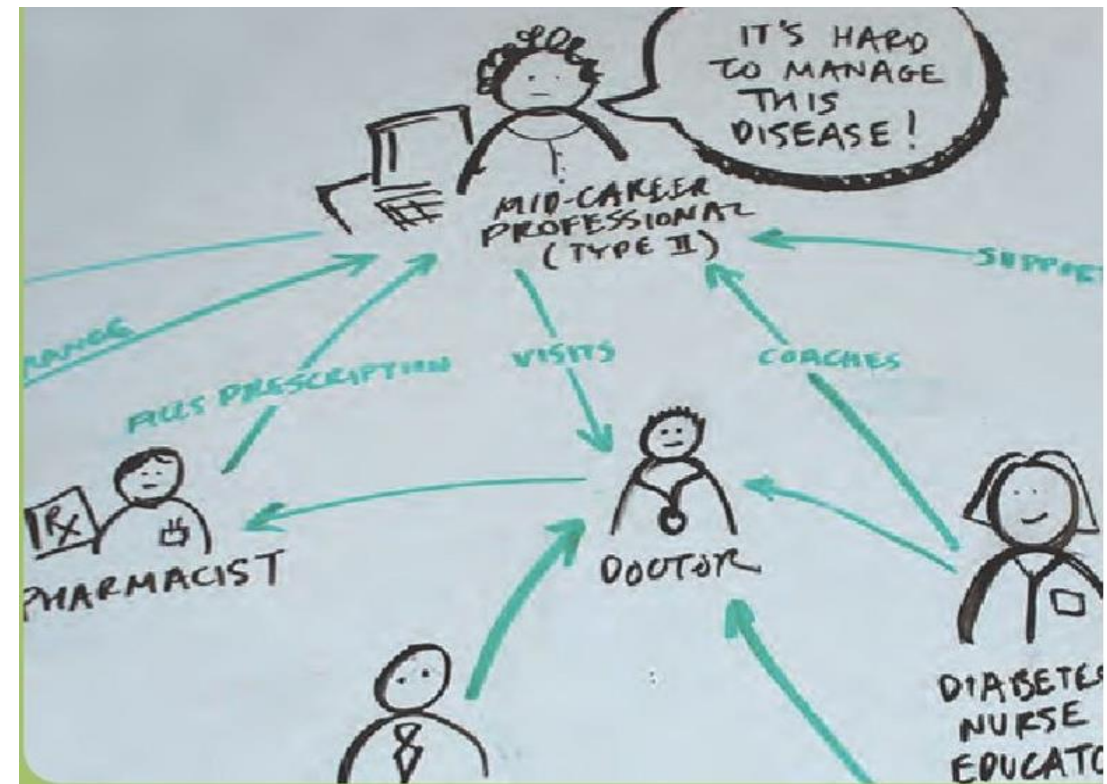
- Understanding **how people interact with systems** (tools, devices, software, etc.) and **identifying the pain points or disconnects in this interaction.**
- Example: Studying how office workers use software to track projects and identifying inefficiencies in its user interface.
- Methods are:
 - Stakeholder Mapping
 - Persona Profile
 - Experience Diagramming
 - Concept Mapping

Stakeholder Mapping

- A way of **diagramming the network of people who have a stake in a given system.**
- The balance of any ecosystem is defined by the **interrelationships of its parts.**
- Each individual unit, whether key or peripheral, plays a role in how the system works.
- As a **visualization of people's relationships, interactions, and needs**, a Stakeholder Map helps you understand the extent and impact of your design decisions.

- In this example, a team was designing new services for people living with chronic diseases. They drafted a full diagram, then decided to focus on professional caregivers.

- Identify a subject area to focus on.
- Convene a diverse team of collaborators.
- Generate a very broad list of stakeholders.
- Draw a symbol of a person for each different type.
- Write a speech-bubble to summarize their mindset.
- Write a label describing their role or title.
- Draw lines with arrows connecting the stakeholders.
- Write a label on the line to describe relationships.
- Circle and label related groupings.



BENEFITS

- Focuses on people above other factors
- Guides your plans for future research
- Documents your research findings
- Builds a shared understanding

Persona Profile

- **An informed summary of the mindset, needs, and goals typically held by key stakeholders.**
- A good set of personas will guide you and your team to think about the recipients of your ideas at every stage of design development.
- **The best Persona Profiles are comprised of illustrations, descriptive text, and a photo portrait.**
- They can help your team remember and discuss the people for whom you are designing.
- **As a reference for generating ideas, prioritizing features, or discussing trade-off decisions, personas are a valuable tool for decision making.**

Here's an example of persona profiles formatted as large posters. They were displayed in the team's workspace as a reminder of the people for whom they were designing.



QUICK GUIDE

- Identify a body of research to inform your work.
- Determine a set of archetypes to develop in detail.
- Write a personal description of each type.
- Give them realistic names.
- Include a representative portrait for each persona.
- Describe their distinguishing characteristics.
- Establish their needs and goals.
- Summarize their mindset with a memorable quote.
- Compose a one-page summary sheet for each type.

HELPFUL HINTS

- Avoid disrespectful stereotypes at all costs.
- Supplement your text with illustrative diagrams.
- Engage an expert designer for visual refinement.

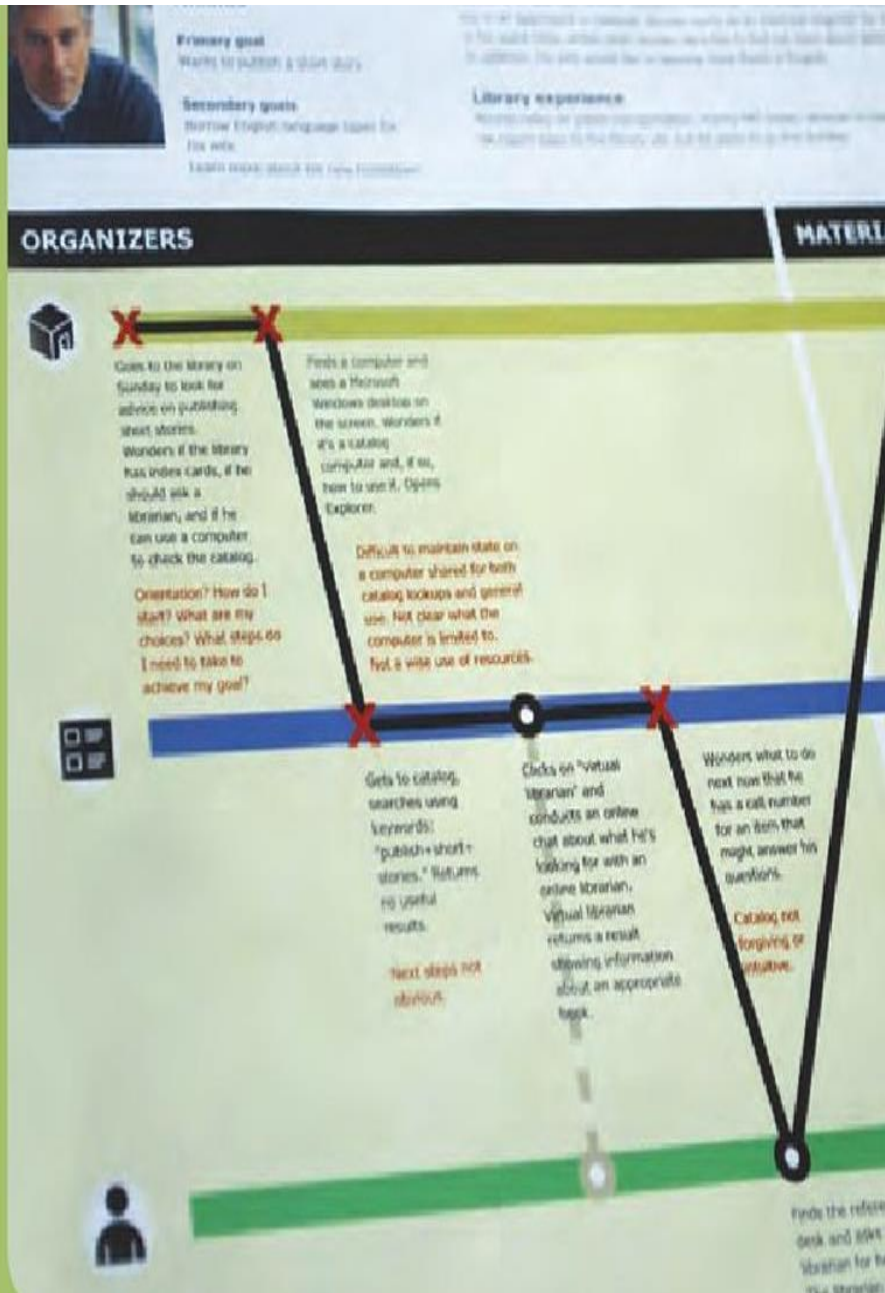
BENEFITS

- Focuses on people above other factors.
- Deepens your empathy for others.
- Summarizes your research findings.
- Challenges your preconceptions.

Experience Diagramming

- **A way of mapping a person's journey through a set of circumstances or task.**
- Experience Diagramming is an **effective way to visualize your event-based research.**
- It is a great way to document the extent to which you understand the current state of a particular situation.
- A useful diagram **reveals more than just an overview of the action.**
- It can also **bring to light important qualities of the experience, showing the complexity or difficulty people faced and how they struggled, adapted, or overcame.**

This example shows a typical path of someone using a large public library system. The design team mapped out the activities and marked opportunities for improvement.



QUICK GUIDE:

- Identify a body of research to inform your work.
- Select an experience to document in detail.
- Focus in on a few key tasks.
- Decide which individuals or personas to represent.
- List the people, places, and things they encounter.
- Determine a format (e.g., flow chart, map, timeline).
- Illustrate a series of typical experiences.
- Highlight the critical waypoints in their journey.

HELPFUL HINTS

- Focus this effort on documenting existing situations.
- Use the diagram to mark typical break points.
- Study the behavioral patterns for potential insights.

BENEFITS

- Summarizes the current state of a situation
- Deepens your empathy for others.
- Documents critical touch points.
- Informs subsequent design activities.

Concept Mapping

- A way of depicting the relationships between various concepts in a given topic area.
- Concept Mapping as a technique for **organizing concepts in a way that illustrates a thorough understanding of a topic, problem, or situation.**
- It **uses diagrams** to illustrate relationships between different concepts, typically by connecting them with labeled arrows.
- Concept maps are widely used for learning, brainstorming, problem-solving, and presenting ideas.

Here's an example of a concept map that was drafted by a team in the music industry. They used the diagram to explore new business models for selling digital music.



- For example, by connecting singers, style, and compositions, it might highlight potential areas for innovation or revenue generation.

QUICK GUIDE:

- Identify a subject area to focus on.
- Form a team of people with multiple perspectives.
- Create a list of concepts related to the topic.
- Think broadly. Include people, places, and things.
- Arrange the concepts in an orderly way.
- Draw lines with arrows to connect related concepts.
- Label the lines with words describing the relations.
- Circle and label related groupings.

HELPFUL HINTS

- Use a very large whiteboard or work surface.
- Put concepts on sticky notes to make them moveable.
- Use the map as a living document.
- Update it often.

BENEFITS

- Deepens knowledge within a subject area.
- Facilitates interdisciplinary collaboration.
- Communicates complex ideas visually.
- Builds a shared understanding.

2. Patterns and Priorities

- Recognizing recurring **patterns** in behavior or processes.
- Understanding **priorities** means identifying what is most important to users in their tasks and activities.
- Example: If users prioritize speed and accuracy in their work, tools and processes can be tailored to support these needs more effectively.
- Methods are:
 - Affinity Clustering
 - Bull's-eye Diagramming
 - Importance/Difficulty Matrix
 - Visualize the Vote

Affinity Clustering / Affinity Diagramming

- A graphic technique for sorting items according to similarity.
- Sometimes in the midst of a project, an over whelming amount of information or ambiguity threatens to bog down the pace of progress.
- Affinity Clustering will help you avoid this road block.
- Whether analyzing research data or considering creative ideas, you can use this method to organize items into logical groups.

In this project, a team was striving to fix problems in a manufacturing facility. They analyzed the various machines and found common areas for improvement.



QUICK GUIDE:

- Identify a topic for consideration.
- Gather a data set (research findings, ideas, etc).
- Record each item on a separate card or sticky note.
- Form a team of collaborators and pick a facilitator.
- Have one person describe, then place, an item.
- Invite others to place similar items in proximity.
- Repeat the pattern until all items are included.
- Discuss and rearrange items as groupings emerge.
- Label the clusters that finally take shape.

HELPFUL HINTS

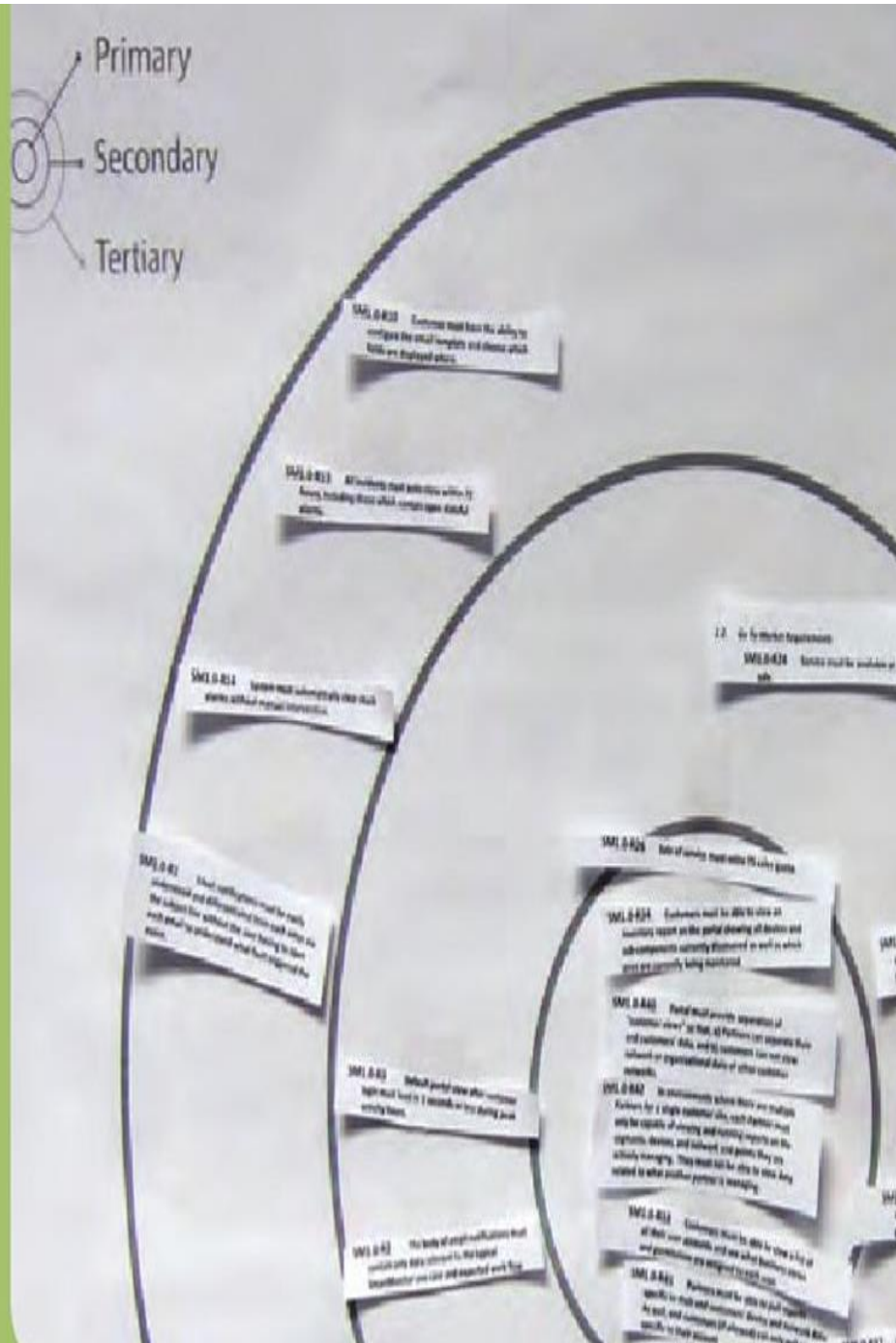
- Don't label the clusters too early. They may shift.
- Look for opportunities to create subgroupings.
- Consider using color to code different types of data

BENEFITS

- Helps you identify issues and insights.
- Reveals thematic patterns.
- Facilitates productive discussion.
- Builds a shared understanding.

Bull's-eye Diagramming

- A way of ranking items in order of importance using a target diagram.
- Bull's-eye Diagram puts a limit on how much you can identify as critical, thus forcing your team to deliberate about essentials before producing anything.
- Since each successive circle is larger than the bull's-eye, you must carefully consider what is critical, what is important, and what is merely peripheral.
- Often this means having to make trade-off decisions.
- The result, however, is a clear delineation of your team's consensus about each item's relative importance.
- It is a comparatively simple method of making difficult decisions.



Here's an example from a team's deliberation about the requirements for a new piece of construction equipment. They used this diagram to prioritize the various features.

QUICK GUIDE:

- Identify a project that requires prioritization.
- Make a large poster with 3 concentric circles.
- Label the circles: Primary, Secondary Tertiary.
- Form a team to conduct an analysis.
- Gather a set of data (e.g., issues, features, etc).
- Divide the data into small units.
- Print small strips of paper or write on sticky notes.
- Debate the relative importance of each item.
- Plot the data on the target, and set priorities.

HELPFUL HINTS

- Size the center ring to fit a limited number of items.
- Enforce a time limit on each round of deliberation.
- Remember that tertiary doesn't mean irrelevant.

BENEFITS

- Helps you determine what is most essential.
- Facilitates productive discussion.
- Builds consensus.
- Helps your team develop a plan of action.

Importance/Difficulty Matrix

- A quad chart for plotting items by relative importance and difficulty.
- A simple 2x2 matrix can be a powerful instrument for establishing priorities.
- Specifically, placing Importance (low to high) on the x-axis and Difficulty (low to high) on the y-axis equips you to work out tensions between these opposing forces.
- When you plot items according to both priorities, you and your team will likely arrive at a workable resolution.
- The items that land in the lower left quadrant are characterized as targeted because they are the easiest to realize.
- The upper left quadrant contains luxurious items—costly endeavors with little return.
- The items in the upper right quadrant are considered to be strategic because they require large investments to get big results.
- The items in the lower right quadrant are high-value because they yield great impact at a low price.

This example shows a team considering new ideas for a start-up company that sells travel supplies. They plotted the ideas and decided which concepts to pursue.



- Identify a project that requires prioritization.
- Make a poster showing a large quad chart.
- Label horizontal axis Importance (or Impact).
- Label vertical axis Difficulty (or Cost to Execute).
- Form a team, and gather data for discussion.
- Plot items horizontally by relative importance.
- Plot items vertically by relative difficulty.
- Consider the quadrants where items get placed.
- Look for related groupings, and set priorities.

HELPFUL HINTS

- Give each item its own place on the relative scale.
- Listen carefully to every point of deliberation.
- Don't view this as a scientific Cost/Benefit study.

BENEFITS

- Helps you prioritize items quickly
- Facilitates deliberation
- Resolves differing opinions
- Helps your team develop a plan of action

Visualize the Vote

- A quick poll of collaborators to reveal preferences and opinions.
- Visualize the Vote is a good way to get everyone's input, giving each person the opportunity to indicate preferences and opinions before final decisions are made.
- Simple method to employ, and offers a good degree of flexibility.
- When using it to pick the best solution among many, you can give everyone a token to cast a single vote.
- Voting provides a quick catalyst for discussion, moving a project toward realization.

In this project, teams presented concepts for attracting young people to a local recreation center. Afterward, everyone involved cast votes for the strongest ideas.



QUICK GUIDE:

- Identify the subject of your polling activity.
- Give each reviewer sticky notes as voting tokens.
- Give each reviewer 1 token to cast an overall vote.
- Give each reviewer 2 tokens to cast as detail votes.
- Announce the criteria for voting.
- Have presenters describe each concept.
- Instruct everyone to vote simultaneously.
- Tally the votes.
- Invite discussion of what people voted for and why.

HELPFUL HINTS

- Use a different colored sticky note for the detail votes.
- Place the token on a specific detail of the concept.
- Consider the cumulative effect of detail votes.

BENEFITS

- Helps you rate and rank preferences
- Reveals thematic patterns
- Diminishes overbearing opinions
- Democratizes decision making

3. Problem Framing

- Involves **defining the problems clearly** based on insights from observations, interviews, and data analysis.
- Problem framing **ensures that the right questions are asked and that the research leads to actionable solutions.**
- Example: Instead of just asking how to improve a software tool, the problem framing might **focus on understanding why users struggle with certain features and what specific tasks need improvement.**
- Methods are:
 - Problem Tree Analysis
 - Statement Starters
 - Abstraction Laddering
 - Rose, Thorn, Bud

Problem Tree Analysis

- A way of exploring the causes and effects of a particular issue.
- Problem Tree Analysis provides a template for mapping causes and effects in order to better understand the chain of connected circumstances that led to the current situation.
- Using the tree as a metaphor, you separate the causes (roots) from the effects (branches) of a central issue (trunk).
- It provides a structured way for your team to reveal concerns, find out causes from symptoms, and potentially frame problem statements in a new and better way.

Here's an example of a team considering the causes and effects of recreational boating accidents. They used this exercise to reframe their subsequent ideation activities.



QUICK GUIDE:

- Identify a problem statement to focus on.
- Assemble a diverse group of stakeholders.
- Gather the team around a flip chart or whiteboard.
- Write your focal problem in the middle of the space.
- Instruct the team to discuss the causes (roots).
- Write the various responses below the focal problem.
- Instruct the team to discuss effects (branches).
- Write the various responses above the focal problem.
- Discuss and decide which cause or effect to focus on.

HELPFUL HINTS

- Acknowledge and discuss direct vs. indirect causes.
- Note that some effects are routine, and some rare.
- Take a quick poll to help decide where to focus.

BENEFITS

- Helps you untangle complex problems
- Reveals various causes and effects
- Builds a shared understanding
- Provides a direction for problem solving

Statement Starters

- An approach to phrasing problem statements that invites broad exploration.
- Statement Starters jump start discussions toward solving the right problem in the best way.
- Statement Starters encourage restating problems as invitations for exploration.
- Statement Starter might convert the challenge into a question like this: “How might we help people listen to music?”
- This phrasing leaves room for multiple interpretations of the challenge, thereby increasing the possibility of finding new and better solutions.

This example shows a lawmaker listing a set of potential statement starters. He asked his constituents to frame their challenge in a new way by filling in the blanks.

How might we —

In what ways might we —

How to —

QUICK GUIDE:

- Identify a set of problems or opportunities.
- State each issue in the form of a short phrase.
- Add a “starter” to the beginning of each phrase.
- Example 1: How might we_____?
- Example 2: In what ways might we_____?
- Example 3: How to_____?
- Pick the best statement starter for each problem.
- Use the new phrasing as a basis for ideation.

HELPFUL HINTS

- Consider adding a “starter” to a research insight.
- Don’t imbed solutions into problem statements.
- Use voting to pick the best problem statement.

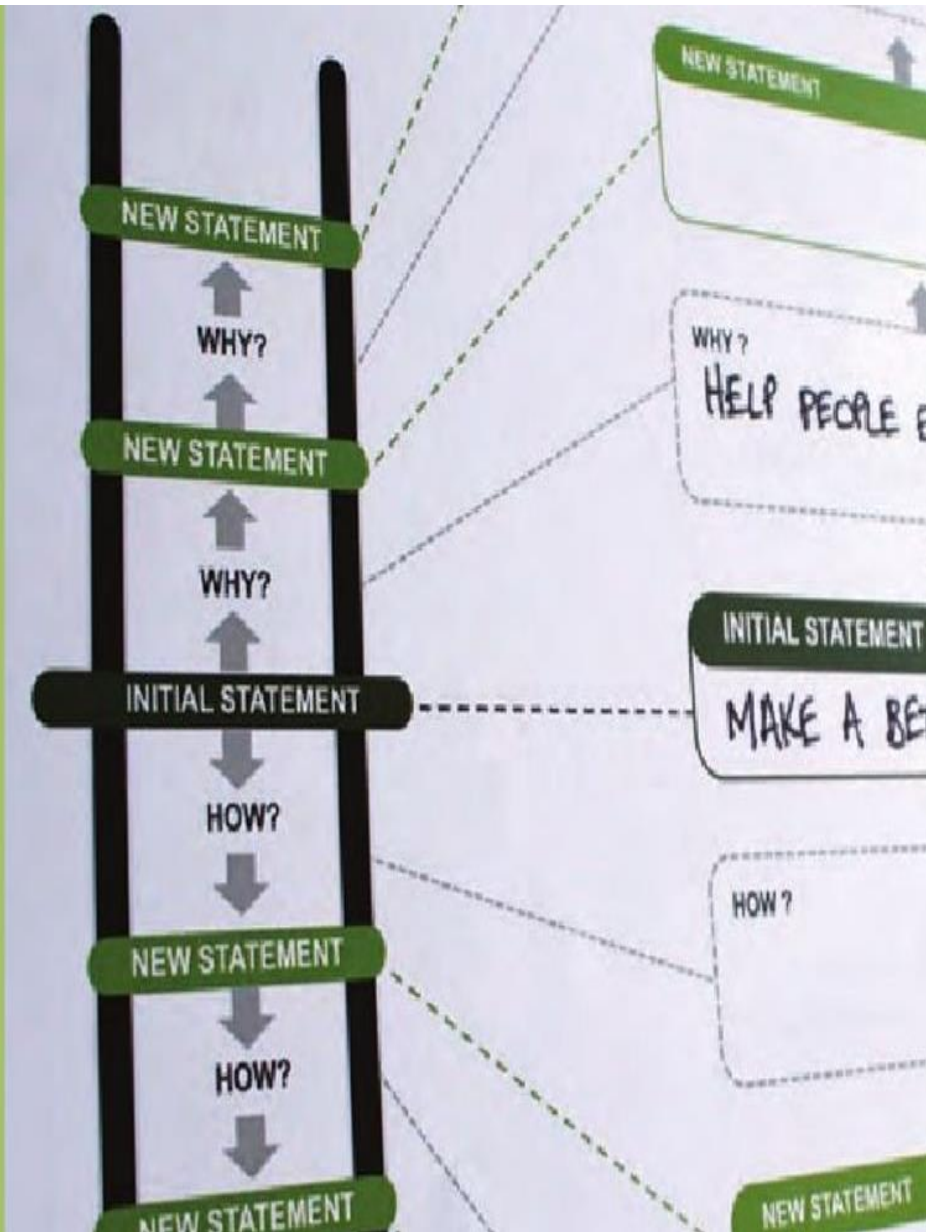
BENEFITS

- Challenges your assumptions
- Helps you see different perspectives
- Provides a direction for problem solving
- Invites divergent thinking

Abstraction Laddering

- A way of reconsidering a problem statement by broadening or narrowing its focus.
- Provides a template for considering a given challenge statement at different levels of focus.
- When you are seeking more visionary solutions, it can help to step back and look at the issue more broadly.
- Moving up the ladder by asking Why? allows you to expand the scope of your inquiry.
- When your scope is so broad that you don't even know how to get started, moving down the ladder helps to tighten your focus.
- By asking How?, the problem is framed more concretely, giving you a more specific challenge to explore.

In this project, the team was challenged to make a better music player for digital songs. They used the ladder to reframe the challenge toward helping people enjoy music.



QUICK GUIDE:

- Identify an initial problem statement.
- Make a laddering worksheet (rungs as blank spaces).
- Write your initial statement on the middle rung.
- Explore various options for revising the statement.
- Move up the ladder by asking Why?
- Consider the options (broader than the initial one).
- Move down the ladder by asking How?
- Consider the options (narrower than the initial one).
- Discuss and decide which option to use for ideation.

HELPFUL HINTS

- Consider the use of Statement Starters to reframe.
- Take a quick poll to help decide which option is best.
- In the end, the initial statement might be the best.

BENEFITS

- Challenges your preconceptions
- Helps you refocus a problem statement
- Builds a shared understanding
- Provides a direction for problem-solving

Rose, Thorn, Bud

- A technique for identifying things as positive, negative, or having potential.
- Adapted for use as a design method, this structure provides an opportunity to analyze a set of data or help scope a problem by revealing focus areas, allowing you to plan next steps.
- You can frame your ensuing activities by documenting observations or opinions on sticky notes as positive, negative, or having potential.
- In all cases, the use of different colors for each note helps you see and consider emergent patterns later on.

This example shows a team member thinking about his company's ability to innovate. The team then used the issue of "Risk" as the basis for an ideation session.



QUICK GUIDE:

- Identify a topic for consideration.
- Assemble a diverse group of stakeholders.
- Give each participant a pen and 3 sticky note pads.
- Explain the topic and the color key.
- Rose = Pink (indicates things that are positive).
- Thorn = Blue (indicates things that are negative).
- Bud = Green (indicates things that have potential).
- Instruct each person to generate many data points.
- Include one issue, insight, or idea per sticky note.

HELPFUL HINTS

- Tell participants to write multiple items per color.
- Resist the temptation to describe solutions here.
- Limit the time frame and the amount of discussion.

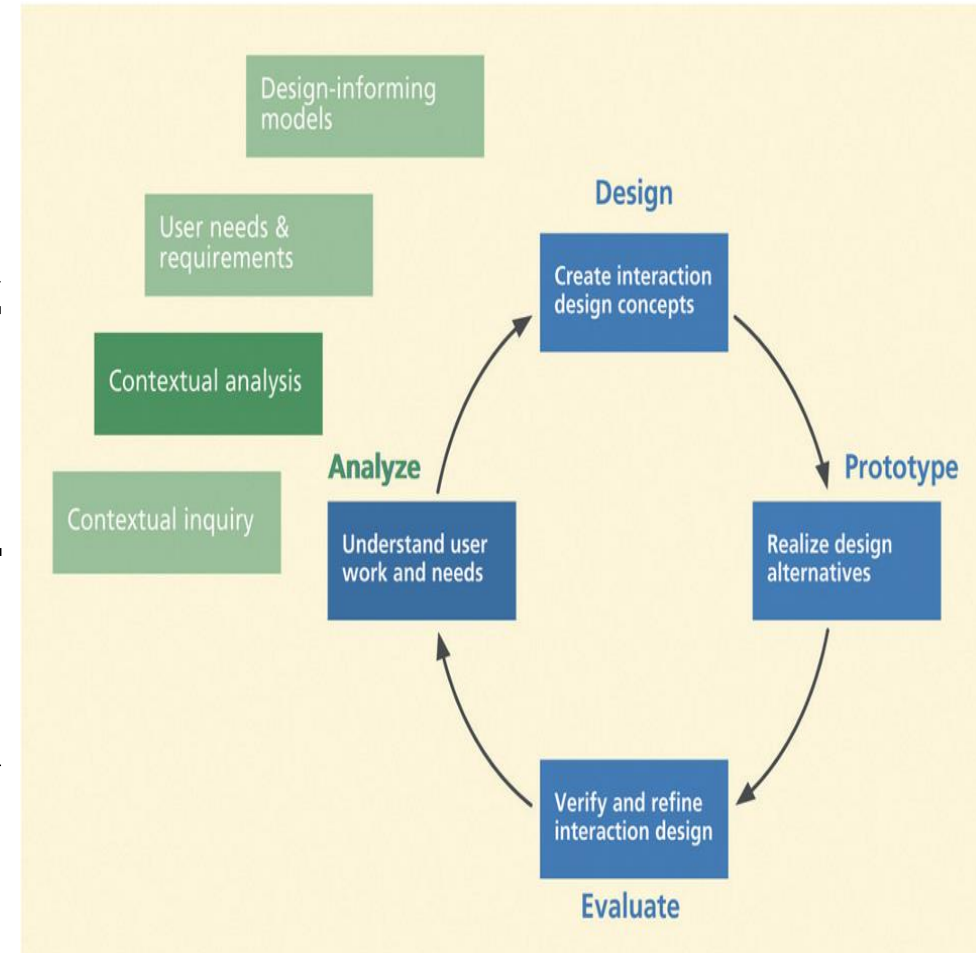
BENEFITS

- Helps you codify research data
- Invites input from all team members
- Facilitates productive discussion
- Helps you identify issues and insights

Contextual Analysis: Consolidating and Interpreting Work Activity Data

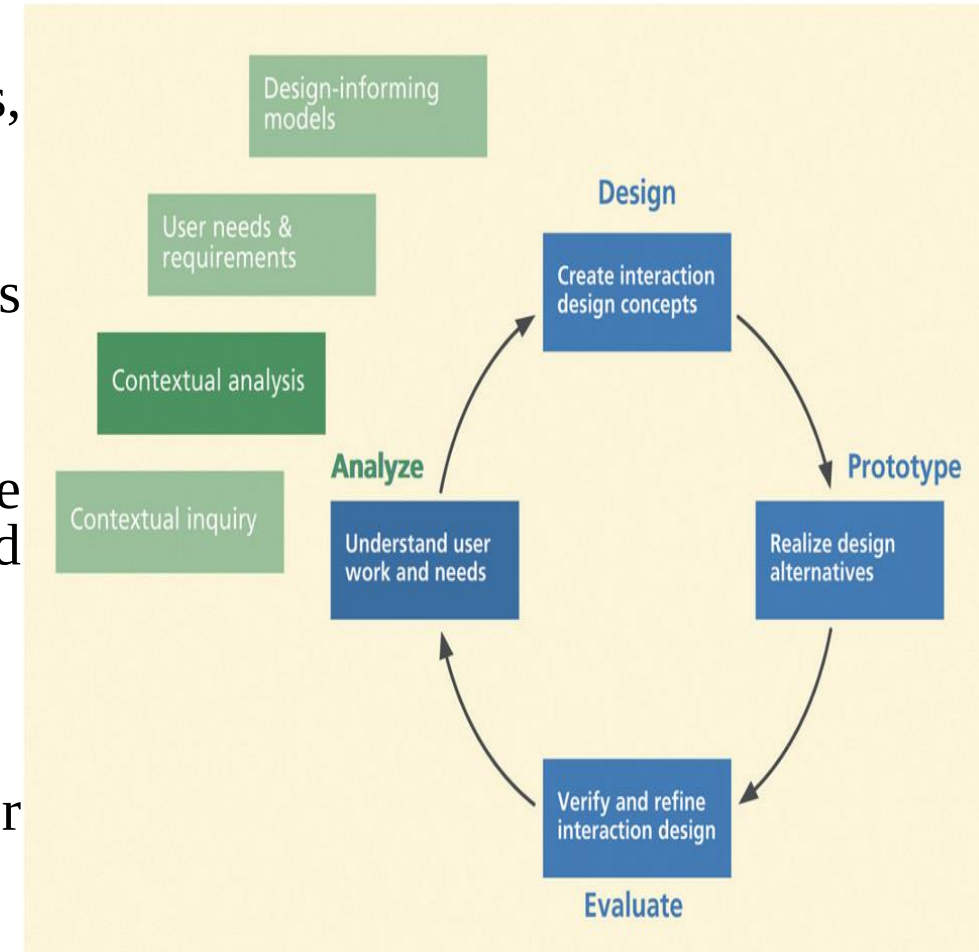
- Once data from contextual inquiries are collected, **contextual analysis** focuses on interpreting and synthesizing this data to draw meaningful conclusions.
- **Consolidating Work Activity Data:**
 - **Data Collection:** Gather qualitative data such as field notes, recordings, or direct observations.
 - **Data Categorization:** Group data into broad themes, tasks, or steps based on user actions, environments, tools used, or challenges encountered.
- **Interpreting Data:**
 - Look for **patterns**, common pain points, and commonalities in how users interact with systems or tools.
 - **Identify Systemic Issues:** Determine if issues are isolated to specific users or systemic across many.
 - **Identify Opportunities for Improvement:** Look for areas where users are inefficient, frustrated, or underutilizing certain features of a system.

- This diagram illustrates a user-centered design (UCD) process that involves iterative phases of **analysis, design, prototyping, and evaluation**.
- **Input Stages**
 - These steps represent the preparatory work and inputs feeding into the UCD process:
 - **Contextual Inquiry**
 - Conduct observations, interviews, and studies in the user's actual work environment to gather qualitative data.
 - **Contextual Analysis**
 - Analyze the collected data to identify patterns, insights, and a deep understanding of user workflows, challenges, and opportunities.
 - **User Needs and Requirements**
 - From the analysis, extract specific user needs and functional requirements that will guide the design.
 - **Design-Informing Models**
 - Develop models such as task flows, personas, and scenarios that describe users and their interaction with the system.



- **Central Iterative Process**

- This is the core UCD cycle, which continuously improves the design based on user feedback and evaluation:
- **Analyze: Understand User Work and Needs**
 - Revisit and refine the understanding of user tasks, goals, and pain points.
- **Design: Create Interaction Design Concepts**
 - Use the insights to develop interaction design concepts that align with user needs and domain requirements.
- **Prototype: Realize Design Alternatives**
 - Build prototypes to explore and visualize alternative designs. These can range from low to highly detailed interactive prototypes.
- **Evaluate: Verify and Refine Interaction Design**
 - Test the prototypes with users to gather feedback.
 - Identify strengths, weaknesses, and areas for improvement.
- **Iterate (Cycle)**
 - Refine the design based on evaluation findings and repeat the process until the design effectively meets user needs.



Key Features of the Process

- **User-Centered Approach**

The focus remains on understanding and addressing user needs throughout the process.

- **Iterative Nature**

The design evolves through cycles of prototyping and evaluation to ensure continuous improvement.

- **Input-Driven**

Insights gained from contextual inquiry and analysis directly inform the design, ensuring it is rooted in real-world user behaviors and requirements.

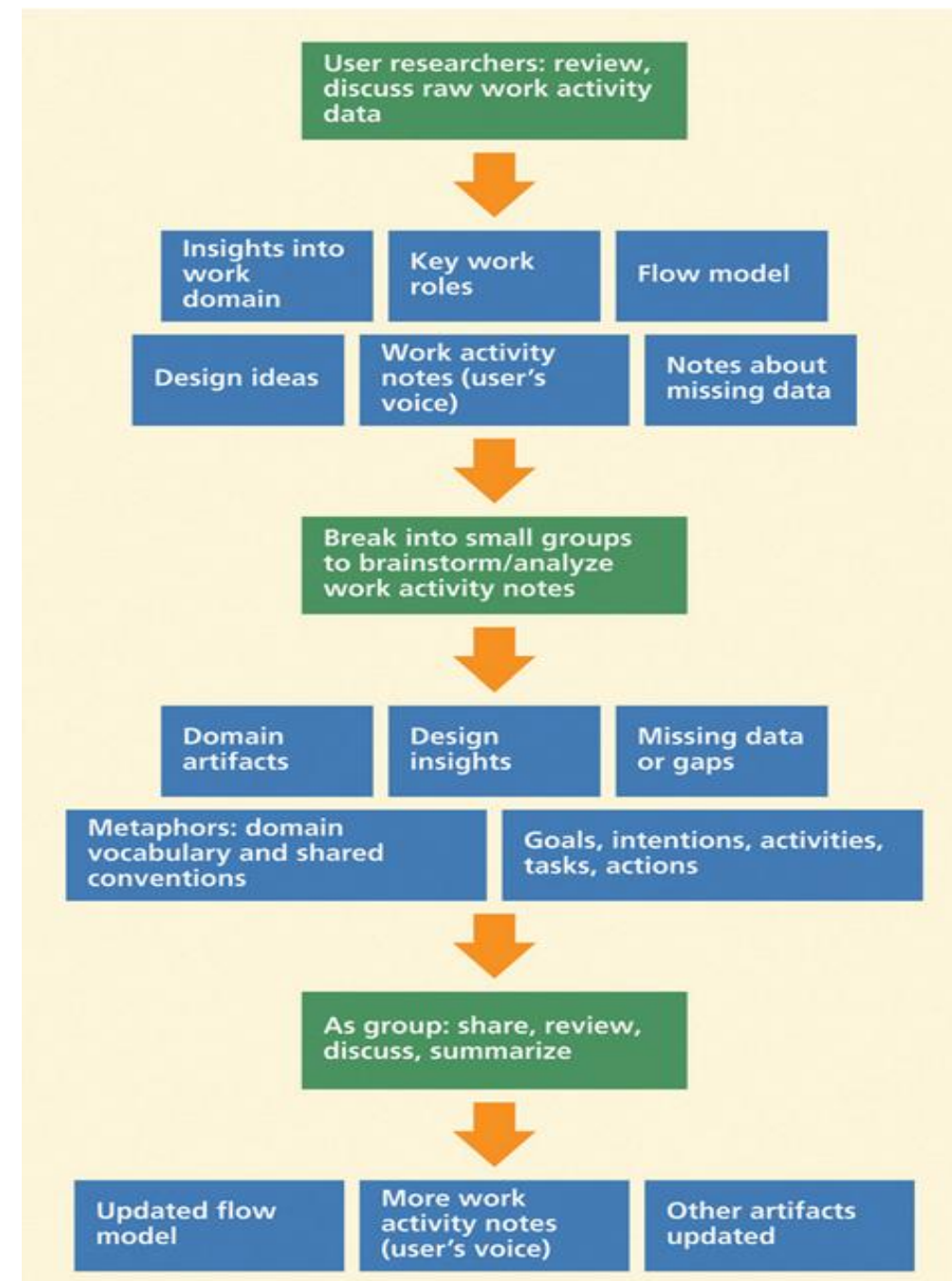
- This model is commonly used in interaction design, UX design, and HCI to create effective, user-friendly systems and products.

Contextual Analysis Is Data Interpretation

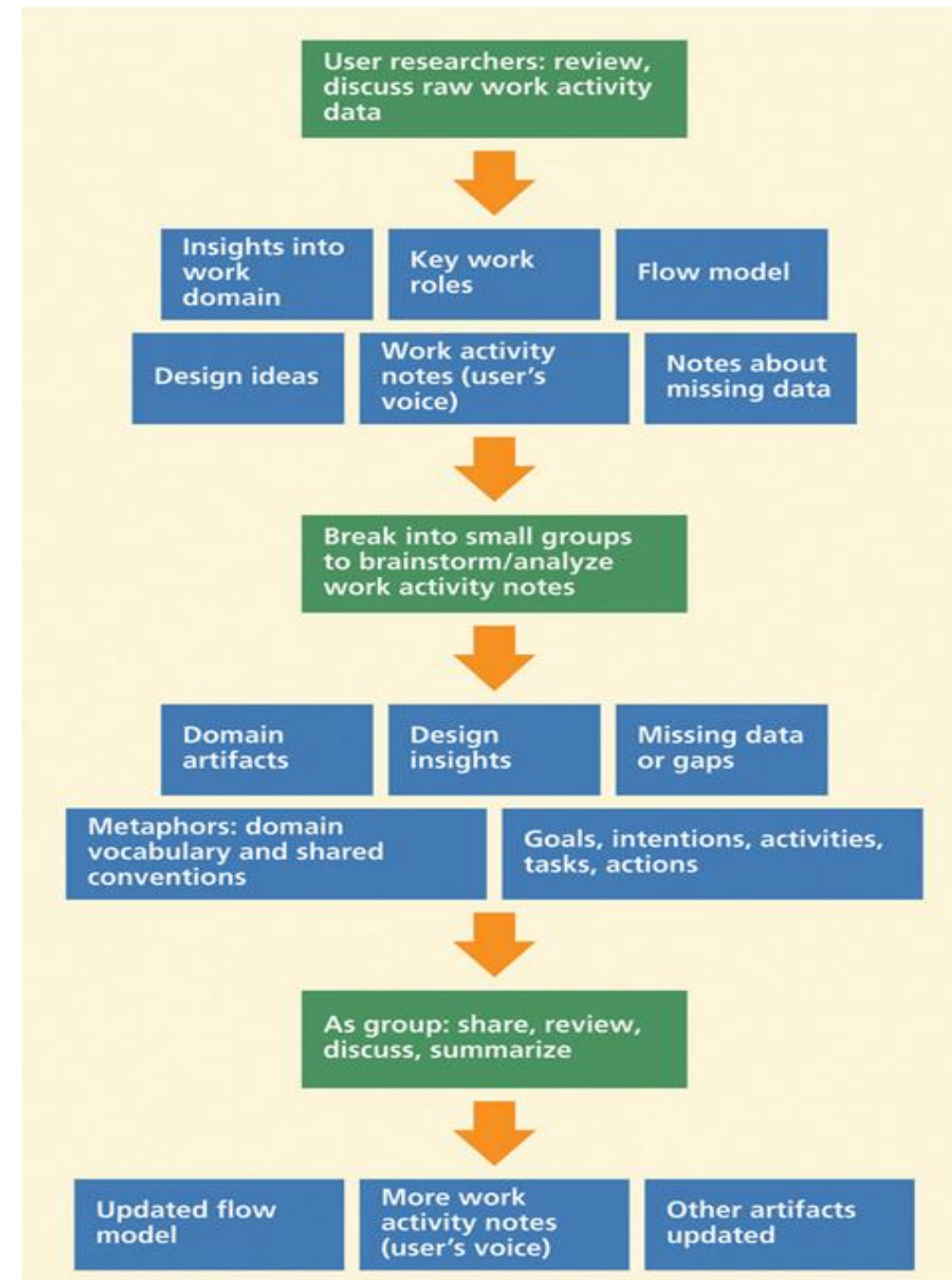
- You have used contextual inquiry to observe and interview users about the nature of their work in context and collected corresponding contextual data, it is now time to analyze that data to understand the work domain.
- **Data consolidation** and communication are accomplished by, respectively:
 - building a work activity affinity diagram (WAAD) from the work activity notes.
 - walkthroughs of all these work products.
- **Interpretation** of raw work activity data is accomplished through:
 - building a flow model and
 - synthesizing work activity notes

Data Interpretation

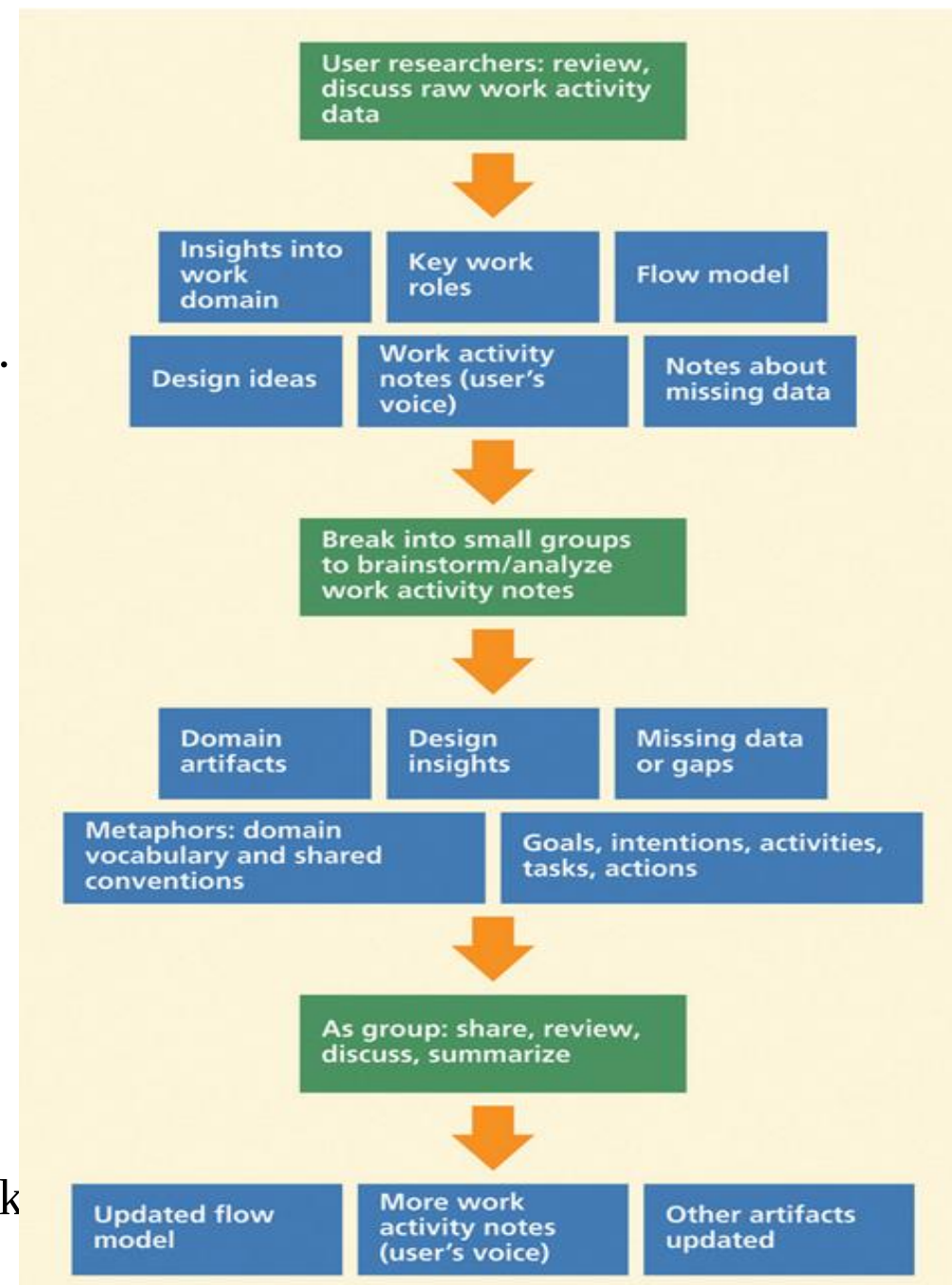
- This flowchart illustrates a structured process for analyzing work activity data in the context of user research and design.
- Here's a step-by-step breakdown of the process:
- **User Researchers: Review and Discuss Raw Work Activity Data (Green Box at the Top)**
 - Researchers gather and analyze raw data about how users perform their work activities.
- **Insights and Notes Generation (First Orange Arrow)**
 - From the raw data, researchers identify key components:
 - **Insights into the work domain:** General understanding of the field or area in question.
 - **Key work roles:** The roles or personas involved in the work.
 - **Flow model:** A high-level representation of how tasks and processes flow.



- **Design ideas:** Early brainstorming ideas for improving or supporting work activities.
- **Work activity notes (user's voice):** Capturing users' input, feedback, and behaviors.
- **Notes about missing data:** Identifying gaps in the information.
- **Small Group Brainstorming/Analysis (Next Orange Arrow)**
 - Teams break into smaller groups to analyze the work activity notes collaboratively.
 - This involves brainstorming and identifying:
 - **Domain artifacts:** Documents, tools, or other items used in the work.
 - **Design insights:** Ideas or patterns relevant to potential solutions.
 - **Missing data or gaps:** Areas requiring further investigation or clarification.



- **Group Outputs (Next Set of Blue Boxes)**
 - The group analysis leads to:
 - **Metaphors:** Establishing shared vocabulary and conventions to describe the domain.
 - **Goals, intentions, activities, tasks, actions:** A deeper understanding of the work structure and user motivations.
- **Group Review, Discussion, and Summarization (Next Green Box)**
 - Teams reconvene as a whole to share their findings, discuss, and summarize insights.
- **Updated Artifacts (Final Outputs, Blue Boxes at the Bottom)**
 - The final outcomes of the process include:
 - **Updated flow model:** A refined representation of task flows.
 - **More detailed work activity notes (user's voice):** Improved notes reflecting deeper understanding.
 - **Other artifacts updated:** Revisions to tools, documents, or other relevant materials.
- This process emphasizes iterative collaboration, with user feedback and work context driving design insights and solutions.



Outputs of Contextual Analysis

- **Work Activity Affinity Diagram (WAAD)**
 - A visual representation of relationships between tasks, roles, tools, and workflows.
 - Used to identify commonalities, variations, and dependencies.
- **Flow Models and Task Models**
 - Flow models show the movement of information, communication, and tasks within the system.
 - Task models break down user tasks into steps, decisions, and goals.
- **Personas and Scenarios**
 - Personas represent archetypes of users based on the analysis.
 - Scenarios describe typical interactions users have with the system.
- **Design Implications and Requirements**
 - A list of actionable design recommendations based on the analysis.
 - Prioritized requirements that address user needs and pain points.
- **Missing Data or Gaps**
 - Areas where additional data collection or validation is needed.

Why Contextual Analysis Matters

- Ensures that design decisions are grounded in real-world user behaviors and needs.
- Reveals opportunities for innovation by identifying unmet needs or inefficiencies.
- Provides a solid foundation for creating user-centered design solutions.
- Helps teams build a shared understanding of the work domain.
- By systematically consolidating and interpreting work activity data, contextual analysis **serves as the bridge between raw user data and actionable design insights.**